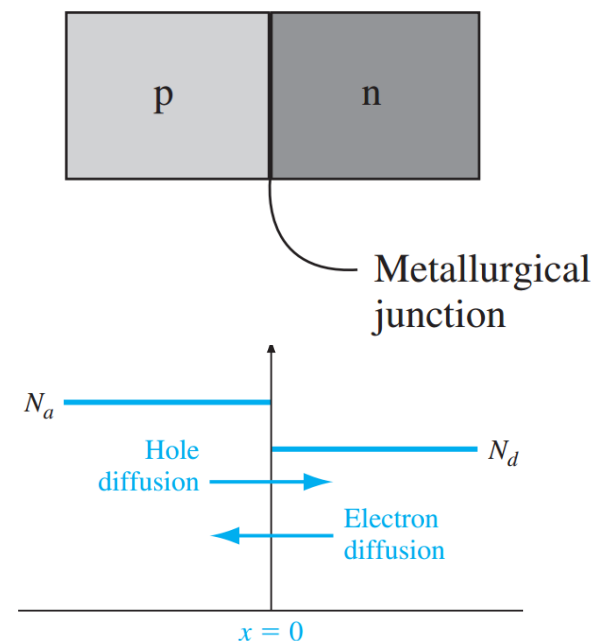


The p-n Junction

By Dr Bhaskar Awadhiya

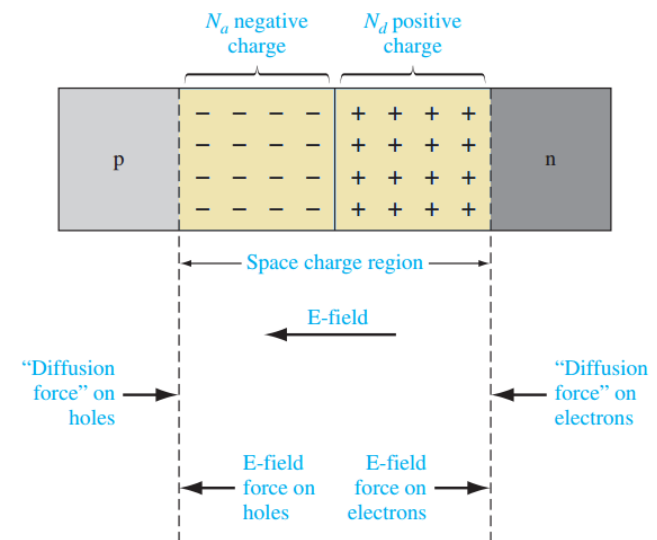
BASIC STRUCTURE OF THE pn JUNCTION

- It is important to realize that the entire semiconductor is a **single-crystal material** in which **one region is doped with acceptor atoms** and the **adjacent region is doped with donor atoms**.
- The interface separating regions is referred to as the **Metallurgical junction**.
- For simplicity, we will consider a **step junction** in which the doping concentration is **uniform**.
- Initially, at the metallurgical junction, there is a very large density gradient in both electron and hole concentrations.
- **Majority carrier electrons** in the n region will **begin diffusing** into the p region.
- **Majority carrier holes** in the p region will **begin diffusing** into the n region.



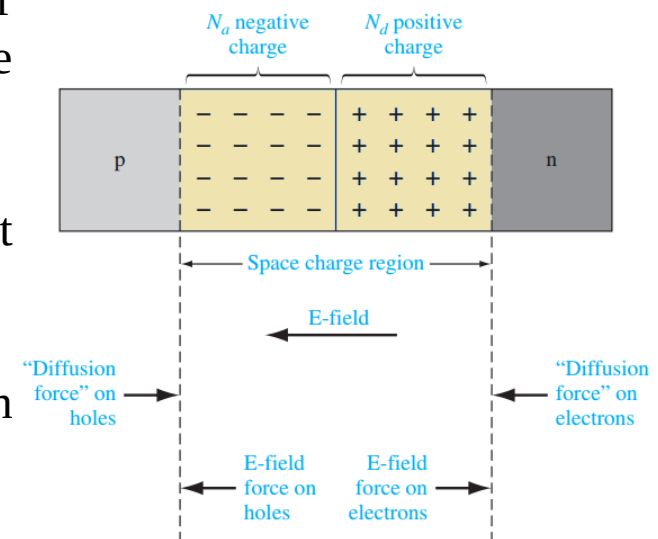
BASIC STRUCTURE OF THE pn JUNCTION

- If we assume there are no external connections to the semiconductor, then this **diffusion process cannot continue indefinitely**.
- As electrons diffuse from the n region, positively **charged donor atoms are left behind**.
- Similarly, as holes diffuse from the p region, they **uncover negatively charged acceptor atoms**.
- The net **positive and negative charges** in the n and p regions **induce an electric field** in the region near the metallurgical junction.
- The **direction of the induced electric field** is from the **positive to the negative charge**, or **from the n to the p region**.



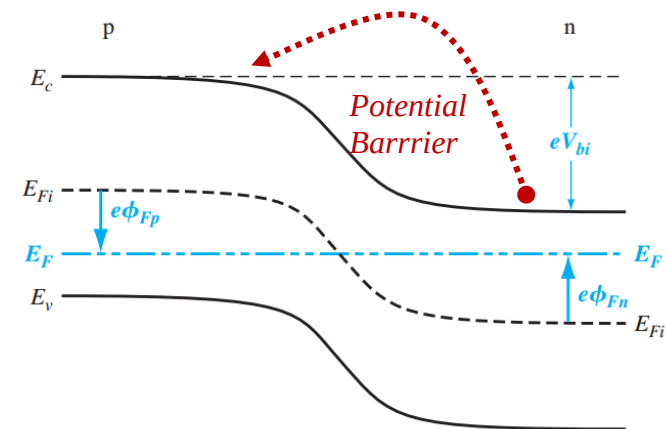
BASIC STRUCTURE OF THE pn JUNCTION

- Regions with charged impurity atoms are referred to as the *space charge region*.
- Essentially all electrons and holes are swept out of the space charge region by the electric field.
- Since the space charge region is **depleted of any mobile charge**, this region is also referred to as the *Depletion region*.
- Density gradients still exist in the majority carrier concentrations at each edge of the space charge region.
- Diffusion forces, acts on the electrons and holes at the edges of the space charge region.
- The **electric field produces another force** which is in the **opposite direction** to the diffusion forces.
- In thermal equilibrium, the diffusion force and the E-field force exactly balance each other.



BUILT-IN POTENTIAL BARRIER

- If we **assume that no voltage is applied** across the pn junction, then the junction is in **thermal equilibrium** —the **Fermi energy level is constant throughout the entire system**.
- Figure shows the energy-band diagram for the pn junction in thermal equilibrium.
- The **conduction and valance band energies must bend** as we go through the space charge region, since the relative position of the conduction and valence bands with respect to the Fermi energy changes between p and n regions.
- Electrons in the conduction band of the n region see a potential barrier in trying to move into the conduction band of the p region.
- This potential barrier is referred to as the **Built-in potential barrier** and is denoted by V_{bi} .



BUILT-IN POTENTIAL BARRIER

- The built-in potential barrier maintains equilibrium between electron in n side (majority) and p side (minority) and also between hole in n side (minority) and p side (majority).
- Is it possible to measure potential barrier using Voltmeter?
- The potential V_{bi} maintains equilibrium, so no current is produced by this voltage.
- The intrinsic Fermi level is equidistant from the conduction band edge through the junction.
- Thus, the built-in potential barrier can be determined as the difference between the intrinsic Fermi levels in the p and n regions.

$$V_{bi} = |e\varphi_{Fn}| + |e\varphi_{Fp}|$$

BUILT-IN POTENTIAL BARRIER

In the n region, the electron concentration in the conduction band is given by

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

which can also be written in the form

$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

where n_i and E_{Fi} are the intrinsic carrier concentration and the intrinsic Fermi energy, respectively. We may define the potential ϕ_{Fn} in the n region as

$$\begin{aligned} e\phi_{Fn} &= E_{Fi} - E_F \\ \Rightarrow n_0 &= n_i \exp\left[\frac{-(e\phi_{Fn})}{kT}\right] \end{aligned}$$

Taking the natural log of both sides of Equation, setting $n_0 = N_d$,

$$\phi_{Fn} = \frac{-kT}{e} \ln\left(\frac{N_d}{n_i}\right)$$

BUILT-IN POTENTIAL BARRIER

Similarly, In the p region

$$\varphi_{Fp} = \frac{+kT}{e} \ln \left(N_a / n_i \right)$$

$$V_{bi} = |\varphi_{Fn}| + |\varphi_{Fp}| = \frac{kT}{e} \ln \left(\frac{N_a N_d}{n_i^2} \right) = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

where $V_t = kT/e$ and is defined as the **Thermal voltage**.

BUILT-IN POTENTIAL BARRIER

OBJECTIVE: Calculate the built-in potential barrier in a pn junction.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 2 \times 10^{17} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$.

The built-in potential barrier is determined as

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right) = (0.0259) \ln \left[\frac{(2 \times 10^{17})(10^{15})}{(1.5 \times 10^{10})^2} \right] = 0.713 \text{ V}$$

If we change the doping concentration in the p region of the pn junction such that the doping concentrations become $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$, then the built-in potential barrier becomes $V_{bi} = 0.635 \text{ V}$

The built-in potential barrier changes only slightly as the doping concentrations change by orders of magnitude because of the logarithmic dependence.

If we change the doping concentration in the p region of the pn junction such that the doping concentrations become $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$, then the built-in potential barrier becomes ?

$$V_{bi} = 0.635 \text{ V}$$

- (a) Calculate the built-in potential barrier in a silicon pn junction at $T = 300 \text{ K}$ for
 (i) $N_a = 5 \times 10^{15} \text{ cm}^{-3}$, $N_d = 10^{17} \text{ cm}^{-3}$ and (ii) $N_a = 2 \times 10^{16} \text{ cm}^{-3}$, $N_d = 2 \times 10^{15} \text{ cm}^{-3}$.
 (b) Repeat part (a) for a GaAs pn junction.

[Ans. (a) (i) 0.736 V, (ii) 0.671 V; (b) (i) 1.20 V, (ii) 1.14 V]

Calculate the built-in potential barrier, V_{bi} , for Si, Ge, and GaAs pn junctions if they each have the following dopant concentrations at $T = 300 \text{ K}$:

(a) $N_d = 10^{14} \text{ cm}^{-3}$	$N_a = 10^{17} \text{ cm}^{-3}$	Si: $V_{bi} = 0.635 \text{ V}$	Si: $V_{bi} = 0.778 \text{ V}$	Si: $V_{bi} = 0.814 \text{ V}$
(b) $N_d = 5 \times 10^{16}$	$N_a = 5 \times 10^{16}$	Ge: $V_{bi} = 0.253 \text{ V}$	Ge: $V_{bi} = 0.396 \text{ V}$	Ge: $V_{bi} = 0.432 \text{ V}$
(c) $N_d = 10^{17}$	$N_a = 10^{17}$	GaAs: $V_{bi} = 1.10 \text{ V}$	GaAs: $V_{bi} = 1.25 \text{ V}$	GaAs: $V_{bi} = 1.28 \text{ V}$

ELECTRIC FIELD

- An electric field is created in the depletion region by the separation of positive and negative space charge densities.
- Figure shows the volume charge density distribution in the pn junction assuming step junction
- The electric field is determined from [Poisson's equation](#), which, for a one dimensional analysis, is

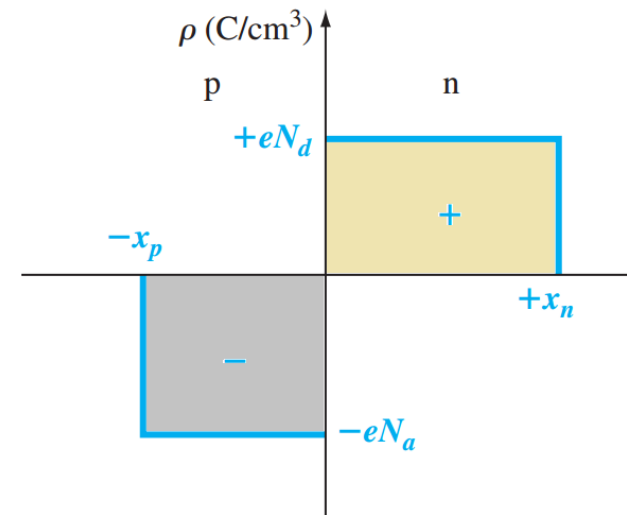
$$\frac{d^2\phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{dE(x)}{dx}$$

where $\phi(x)$ is the electric potential, $E(x)$ is the electric field, $\rho(x)$ is the volume charge density, and ϵ_s is the permittivity of the semiconductor

- From Figure the charge densities (charge/volume) are

$$\rho(x) = -eN_a \quad -x_p < x < 0$$

$$\rho(x) = eN_d \quad 0 < x < x_n$$



ELECTRIC FIELD

- The electric field in the p region is found by poisson's equation as

$$E = \int \frac{\rho(x)}{\epsilon_s} dx = \int \frac{-eN_a}{\epsilon_s} dx = \frac{-eN_a}{\epsilon_s} x + C_1$$

- The electric field is assumed to be zero in the neutral p region for $x < -x_p$ since the currents are zero in thermal equilibrium.
- Since there are no surface charge densities within the pn junction structure, the electric field is a continuous function.
- The constant of integration is determined by setting $E=0$ at $x=-x_p$. The electric field in the p region is then given by

$$E = \frac{-eN_a}{\epsilon_s} (x + x_p) \quad -x_p \leq x \leq 0$$

- In the n region, the electric field is determined from

$$E = \int \frac{eN_d}{\epsilon_s} dx = \frac{eN_d}{\epsilon_s} x + C_2$$

ELECTRIC FIELD

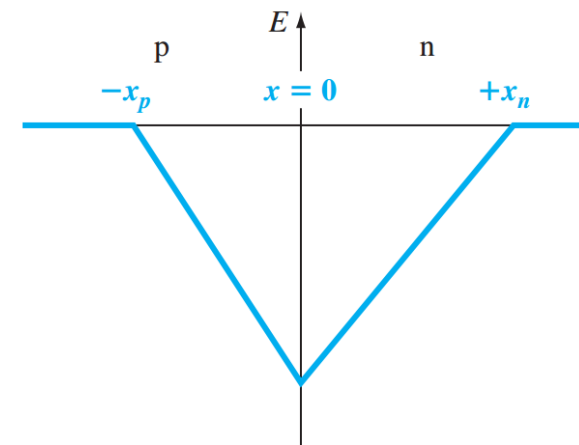
- C_2 is again a constant of integration and is determined by setting $E=0$ at $x = x_n$, since the E-field is assumed to be zero in the n region and is a continuous function. Then

$$E = \frac{-eN_d}{\epsilon_s} (x_n - x) \quad 0 \leq x \leq x_n$$

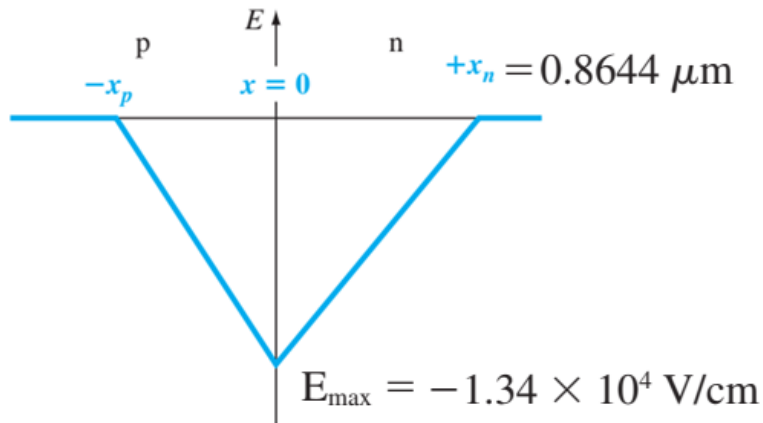
- The electric field is also continuous at the metallurgical junction, or at $x = 0$.

$$N_a x_p = N_d x_n$$

- Above equation states that the number of negative charges per unit area in the p region is equal to the number of positive charges per unit area in the n region.
- The electric field direction is from the n to the p region, or in the negative x direction for this geometry.
- For the uniformly doped pn junction, the E-field is a linear function of distance through the junction.

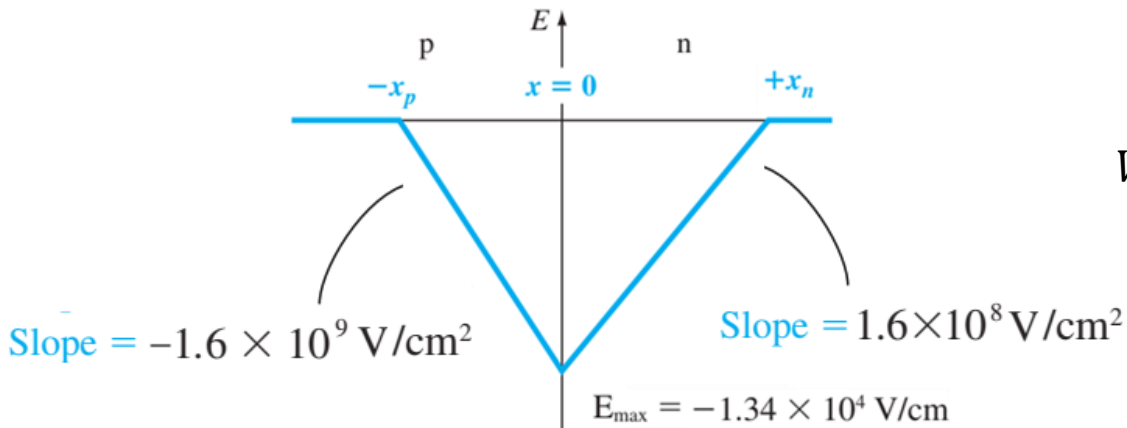


The electric profile for silicon p-n junction diode at $T=300\text{K}$ is shown below, It has a depletion width of $0.951\mu\text{m}$. Calculate the built-in potential.



$$V_{bi} = 0.635 \text{ V.}$$

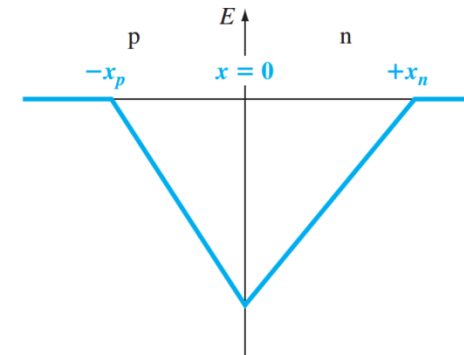
The electric profile for silicon p-n junction diode at $T=300\text{K}$ is shown below, Calculate the built-in potential and the depletion width.



$$V_{bi} = 0.6368\text{V and } W = 0.92125\mu\text{m}$$

POTENTIAL

- The **maximum (magnitude)** electric field occurs **at the metallurgical junction**.
- An **electric field exists** in the depletion region **even when no voltage is applied** between the p and n regions.
- The potential in the junction is found by integrating the electric field. In the p region then, we have



$$\phi(x) = -\int E(x)dx = \int \frac{eN_a}{\epsilon_s}(x + x_p)dx$$

$$\phi(x) = \frac{eN_a}{\epsilon_s} \left(\frac{x^2}{2} + x_p \cdot x \right) + C_1$$

- The potential difference through the pn junction is the important parameter, rather than the absolute potential.
- We may arbitrarily set the potential equal to zero at $x=x_p$. The constant of integration is then found as

BUILT IN POTENTIAL

$$C_1' = \frac{eN_a}{2\epsilon_s} x_p^2$$

so that the potential in the p region can now be written as

$$\phi(x) = \frac{eN_a}{2\epsilon_s} (x + x_p)^2 \quad (-x_p \leq x \leq 0)$$

We determine the potential in the n region by integrating the electric field

$$\phi(x) = \int \frac{eN_d}{\epsilon_s} (x_n - x) dx$$

$$\phi(x) = \frac{eN_d}{\epsilon_s} \left(x_n \cdot x - \frac{x^2}{2} \right) + C_2'$$

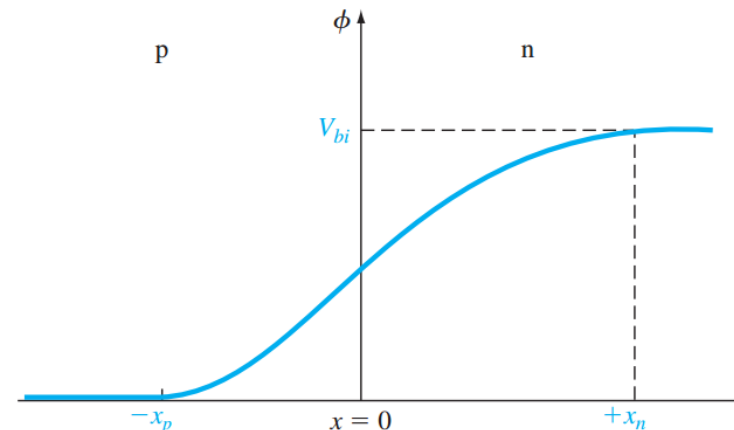
The potential is a continuous function, so at the metallurgical junction, or at $x=0$, gives

$$C_2' = \frac{eN_a}{2\epsilon_s} x_p^2$$

$$\phi(x) = \frac{eN_d}{\epsilon_s} \left(x_n \cdot x - \frac{x^2}{2} \right) + \frac{eN_a}{2\epsilon_s} x_p^2 \quad (0 \leq x \leq x_n)$$

POTENTIAL

- Figure is a plot of the potential through the junction and shows the quadratic dependence on distance.
- The magnitude of the potential at $x = x_n$ is equal to the built in potential barrier.
- Then we have



$$V_{bi} = |\phi(x = x_n)| = \frac{e}{2\epsilon_s} (N_d x_n^2 + N_a x_p^2)$$

- The potential energy of an electron is given by $E = e\phi$, which means that the electron potential energy also varies as a quadratic function of distance through the space charge region.

SPACE CHARGE WIDTH

We can determine the distance that the space charge region extends into the p and n regions from the metallurgical junction. This distance is known as the space charge width.

$$N_a x_p = N_d x_n \Rightarrow x_p = \frac{N_d x_n}{N_a}$$

Then, substituting in the V_{bi} equation and solving for x_n , we obtain

$$x_n = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a}{N_d} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2}$$

Above equation gives the space charge width, or the width of the depletion region, x_n extending into the n-type region for the case of zero applied voltage.

Similarly solving for x_p

$$x_p = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_d}{N_a} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2}$$

where x_p is the width of the depletion region extending into the p region for the case of zero applied voltage.

SPACE CHARGE WIDTH

The total depletion or space charge width W is the sum of the two components, or

$$W = x_p + x_n$$

$$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$$

OBJECTIVE: Calculate the space charge width and electric field in a pn junction for zero bias.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$.

In previous example we determined the built-in potential barrier as $V_{bi} = 0.635$ V.

The space charge width is

$$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2} = \left\{ \frac{2(11.7)(8.85 \times 10^{-14})(0.635)}{1.6 \times 10^{-19}} \left[\frac{10^{16} + 10^{15}}{10^{16} \times 10^{15}} \right] \right\}^{1/2}$$

$$W = 0.951 \times 10^{-4} \text{ cm} = 0.951 \mu\text{m}$$

SPACE CHARGE WIDTH

We can also find x_n and x_p as $x_n = 0.8644 \mu\text{m}$, and $x_p = 0.0864 \mu\text{m}$.

The peak electric field at the metallurgical junction is,

$$E = \frac{-eN_d}{\epsilon_s}(x_n) = -\frac{(1.6 \times 10^{-19})(1 \times 10^{15})(0.8644 \times 10^{-4})}{(11.7)(8.85 \times 10^{-14})} = -1.34 \times 10^4 \text{ V/cm}$$

- The peak electric field in the space charge region of a pn junction is quite large.
- We must keep in mind, however, that there is **no mobile charge in this region; hence there will be no drift current.**
- We may also note, that **the width of each space charge region is a reciprocal function of the doping concentration**
- The depletion region will extend further into the lower-doped region.

A silicon pn junction at $T = 300$ K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 5 \times 10^{15} \text{ cm}^{-3}$. Determine x_n , x_p , W , and $|E_{\max}|$.

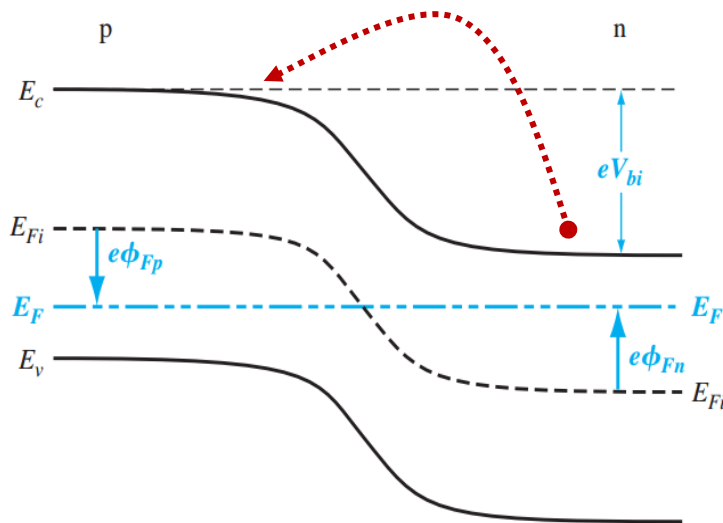
$$\begin{aligned} & \text{(Ans. } x_n = 4.11 \times 10^{-6} \text{ cm,} \\ & x_p = 4.11 \times 10^{-5} \text{ cm, } W = 4.52 \times 10^{-5} \text{ cm, } |E_{\max}| = 3.18 \times 10^4 \text{ V/cm)} \end{aligned}$$

Calculate V_{bi} , x_n , x_p , W , and $|E_{\max}|$ for a silicon pn junction at zero bias and $T = 300$ K for doping concentrations of (a) $N_a = 2 \times 10^{17} \text{ cm}^{-3}$, $N_d = 10^{16} \text{ cm}^{-3}$ and (b) $N_a = 4 \times 10^{15} \text{ cm}^{-3}$, $N_d = 3 \times 10^{16} \text{ cm}^{-3}$.

$$\begin{aligned} & \text{[Ans. (a) } V_{bi} = 0.772 \text{ V, } x_n = 0.3085 \text{ } \mu\text{m, } x_p = 0.0154 \text{ } \mu\text{m, } W = 0.3240 \text{ } \mu\text{m,} \\ & |E_{\max}| = 4.77 \times 10^4 \text{ V/cm; (b) } V_{bi} = 0.699 \text{ V, } x_n = 0.0596 \text{ } \mu\text{m, } x_p = 0.4469 \text{ } \mu\text{m,} \\ & W = 0.5064 \text{ } \mu\text{m, } |E_{\max}| = 2.76 \times 10^4 \text{ V/cm}] \end{aligned}$$

An abrupt Si p-n junction has $N_a = 10^{18} \text{ cm}^{-3}$ on one side and $N_d = 5 \times 10^{15} \text{ cm}^{-3}$ on the other.

- Calculate the Fermi level positions at 300 K in the p and n regions.
- Draw an equilibrium band diagram for the junction and determine the contact potential V_0 from the diagram.



$$n_0 = n_i \exp \left[\frac{E_F - E_{Fi}}{kT} \right] \quad p_0 = n_i \exp \left[\frac{E_{Fi} - E_F}{kT} \right]$$

$$E_F - E_{Fi} = kT \ln \frac{n_0}{n_i} \quad E_F - E_{Fi} = 0.329 \text{ eV}$$

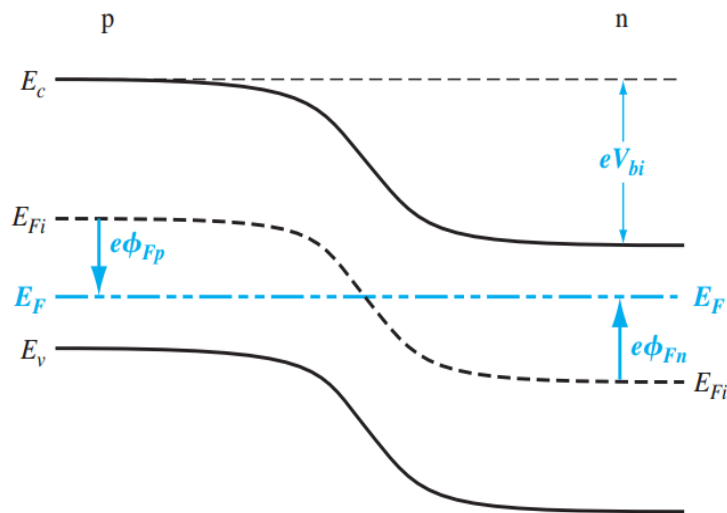
$$E_{Fi} - E_F = kT \ln \frac{p_0}{n_i} \quad E_{Fi} - E_F = 0.467 \text{ eV}$$

$$eV_{bi} = e|\phi_{Fn}| + e|\phi_{Fp}| \quad eV_{bi} = 0.467 \text{ eV} + 0.329 \text{ eV} = 0.796 \text{ eV}$$

$$eV_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right) = 0.796 \text{ eV}$$

A silicon pn junction in thermal equilibrium at $T = 300$ K is doped such that $E_F - E_{Fi} = 0.365$ eV in the n region and $E_{Fi} - E_F = 0.330$ eV in the p region.

(a) Sketch the energy-band diagram for the pn junction. (b) Find the impurity doping concentration in each region. (c) Determine V_{bi} .



$$eV_{bi} = e|\phi_{Fn}| + e|\phi_{Fp}|$$

$$eV_{bi} = 0.467 \text{ eV} + 0.329 \text{ eV} = 0.796 \text{ eV}$$

$$eV_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right) = 0.796 \text{ eV}$$

An abrupt silicon pn junction at zero bias has dopant concentrations of $N_a = N_d = 2 \times 10^{16} \text{ cm}^{-3}$, $T = 300 \text{ K}$. (a) Calculate the Fermi level on each side of the junction with respect to the intrinsic Fermi level. (b) Sketch the equilibrium energy- band diagram for the junction and determine V_{bi} . (c) Determine x_n , x_p , and the peak electric field for this junction.

- (a) n side: $E_F - E_{Fi} = 0.3653 \text{ eV}$,
p side: $E_{Fi} - E_F = 0.3653 \text{ eV}$;
(b) $V_{bi} = 0.7306 \text{ V}$;
(c) $x_n = 0.154 \mu\text{m}$, $x_p = 0.154 \mu\text{m}$,
 $|E_{\text{max}}| = 4.75 \times 10^4 \text{ V/cm}$